EXERCISE 2

ROLL NO:240701248 NAME:KAVYASRI M

Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION PAGE

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that organizes information into smaller, meaningful units, helping users understand and remember instructions more effectively. The instruction page of the **Neuroquest cosmic memory challenge** applies chunking principles in the following ways:

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1. Clear and Sequential Numbering

- The instructions are divided into six numbered steps, allowing users to process the task one step at a time.
- Each step communicates only one action, preventing confusion and reducing mental load.
- The logical flow ensures users know exactly what to do next.

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2. Logical Grouping of Information

- Introduction Phase (Step 1) → Briefly explains the purpose of the memory challenge.
- Observation Phase (Steps 2–3) → Informs users about viewing emojis arranged in a grid.
- Memorization Phase (Step 4) → Encourages users to focus on remembering positions and patterns.
- Recall Phase (Step 5) → Explains how users must recall the emojis after they disappear.
- Performance Focus (Step 6) → Emphasizes accuracy over speed for better results.

This grouping helps users mentally prepare for each stage of the task.

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3. Visual Hierarchy and Design Elements

- A bold and prominent title ("MEMORY CHALLENGE") immediately communicates the purpose of the screen.
- Proper spacing and bullet-style formatting improve readability and reduce visual clutter.

- A clearly visible START button guides users toward the next interaction.
- Emoji-based and space-themed visuals make the interface engaging while maintaining clarity.

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4. Time Constraint Reinforcement

- The instruction page clearly states that users will have limited time to view the grid.
- This sets expectations in advance and encourages focused attention during the memorization phase.
- Explicit timing information reduces uncertainty and anxiety.

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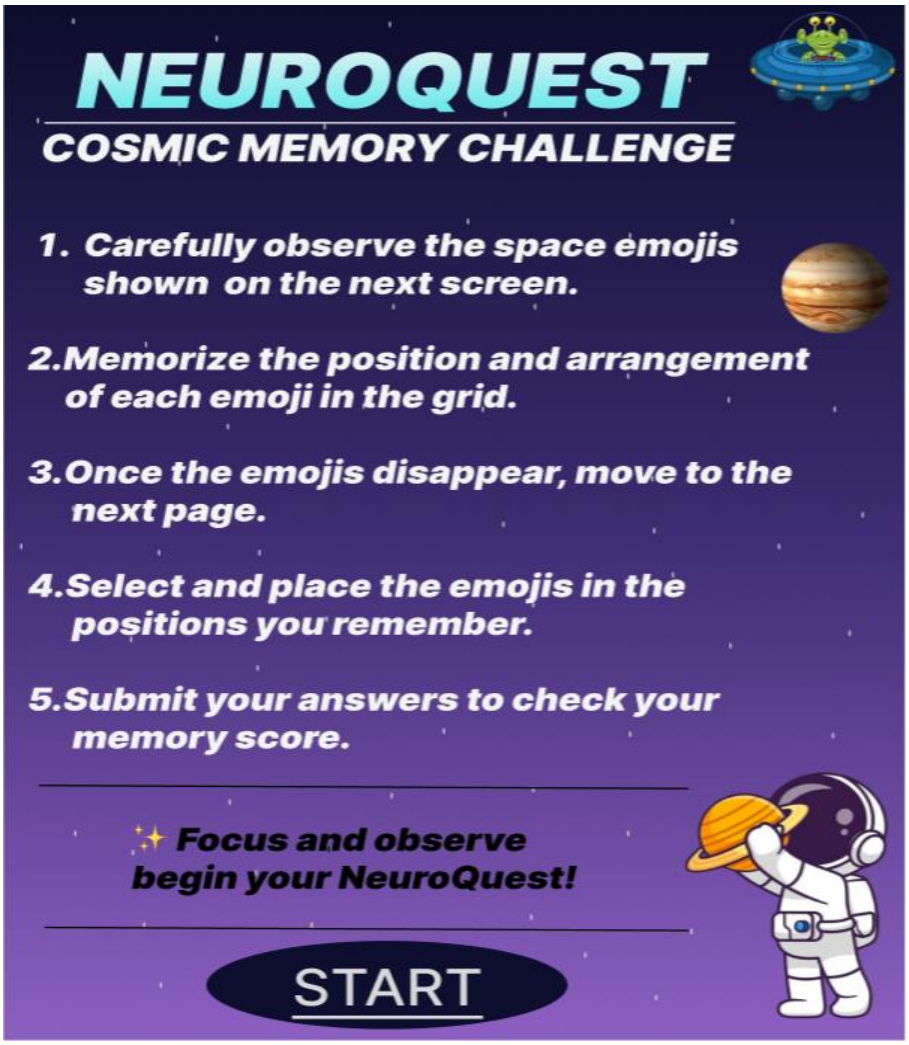
5. Simplicity and Clarity

- Instructions use short, simple sentences that are easy to understand.
- Action-oriented language keeps users engaged and confident.
- No unnecessary technical terms are used, making the task accessible to all users.



FRAME 2: CHUNKING PHASE

Analysis of the cosmic memory Task – Chunking Phase Screen

This screen represents the Chunking Phase of the cosmic memory Task, where users observe and mentally organize a limited set of emojis within a short time frame. The design encourages users to apply chunking strategies to improve recall accuracy.

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1. Purpose of the Screen

- This screen functions as the memory encoding stage, where users view and memorize emoji positions.
- The term *Chunking Phase* indicates the use of the chunking principle, which helps users remember information by grouping related items.
- Users are encouraged to identify patterns and associations among emojis rather than memorizing each item individually.

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2. Key Elements and UI Components

⏳ Countdown Timer (Top Corner – "00:15")

- Displays the 15-second time limit for memorization.
- The contrasting color and bold font create a sense of urgency.
- Helps users stay focused and aware of the limited observation time.

⏱ Time Progress Indicator

- Visually represents the remaining time.
- Reinforces time awareness without requiring users to read the timer constantly.
- Supports quick decision-making and attention management.

🌀 Cosmic Emoji Grid (Central Area)

- A structured grid containing 9 visually distinct emojis.
- Emojis are evenly spaced to reduce visual clutter.
- The layout supports chunking strategies such as:
    - Grouping by emoji type (faces, objects, symbols).
    - Grouping by color tones or visual similarity.
    - Remembering spatial patterns (corners, center, rows).

✨ Minimal Thematic Background

- A calm, space-inspired background enhances focus without distraction.
- Supports immersion while keeping attention on the emojis.
- Maintains visual balance and readability.

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3. How the Chunking Phase Works

1. Users observe the emoji grid as soon as the screen appears.
2. They mentally group emojis using patterns, positions, or similarities.
3. The countdown timer restricts viewing time, encouraging efficient memorization.
4. Once the timer ends, users are automatically prepared for the recall phase.

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4. Cognitive and UX Benefits of Chunking

- Improves short-term memory performance by enabling grouped recall.
- Reduces cognitive overload by organizing information visually.
- Enhances pattern recognition and spatial memory.
- Creates a smooth transition from observation to recall, improving overall task effectiveness.

Memorize the Cosmic Grid

FRAME 3: RECALL PHASE

Analysis of the cosmic memory Task – Selection Phase Screen

This screen represents the Recall Phase of the cosmic memory Task, where users retrieve information from short-term memory and select the emojis they remember from the previous Chunking Phase.

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1. Purpose of the Screen

- This screen functions as the spatial memory retrieval stage, where users recall both the identity and position of emojis.
- It evaluates how effectively users encoded spatial information during the Chunking Phase.
- The task emphasizes accuracy of placement, not speed.

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2. Key Elements and UI Components

🌍 Title & Instruction Text

- "PLACE THE EMOJIS" (Bold, uppercase title)
    - Clearly communicates the main task objective.
    - Visually prominent to grab attention.
- "DRAG THE EMOJIS TO THE CORRECT SPOT"
    - Simple, action-oriented instruction.
    - Reduces confusion and guides user interaction clearly.

▯ Empty 3×3 Grid (Recall Grid)

- A 3×3 grid with empty slots represents the recall area.
- Users must reconstruct the emoji positions they memorized earlier.
- The grid layout:
    - Encourages spatial chunking (rows, columns, corners, center).
    - Helps users recall emojis by grouping positions mentally.

🚀 Emoji Selection Panel (Bottom Tray)

- A horizontal tray contains nine space-themed emojis, such as:

    - Rocket, alien, planet, moon, star, astronaut, UFO, galaxy.
- This panel includes:
    - Emojis shown during the Chunking Phase.
    - Optional distractor emojis to increase difficulty.
- Users drag emojis from this panel into the grid, reinforcing active recall.

▶ “NEXT” Button

- Positioned at the bottom for clear navigation.
- Allows users to proceed only after completing placements.
- Signals transition to the results or feedback phase.

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3. How the Recall Phase Works

1. Users view the empty grid and recall the earlier emoji arrangement.
2. They drag emojis from the selection tray into the grid.
3. Users mentally rely on spatial memory patterns (rows, symmetry, clusters).
4. Once satisfied, users click “NEXT” to submit their placement.
5. The system evaluates placement accuracy in the next screen.

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4. Cognitive and UX Benefits

- Tests spatial memory along with visual recall.
- Encourages deeper memory encoding compared to simple multiple-choice selection.
- Drag-and-drop interaction increases engagement and user involvement.
- Grid-based recall supports chunking strategies like:
    - Remembering corners first.
    - Grouping emojis by rows or shapes.
- Minimal distractions keep cognitive load low.

🌍 PLACE THE EMOJIS

DRAG THE EMOJIS TO THE CORRECT SPOT

NEXT

FRAME 4: RESULT PAGE

Analysis of the Memory Recall Task – Score & Feedback Screen

This screen represents the **Score & Feedback Phase** of the NeuroQuest Memory Recall Task, where users receive immediate feedback on their memory performance based on their emoji placement accuracy in the previous Recall Phase.

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1. Purpose of the Screen

- Provides **instant feedback** on the user's memory recall accuracy.
- Displays performance results in a clear and motivating manner.
- Allows users to decide their next course of action after viewing their score.

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2. Key Elements & UI Components

🎯 Title & Score Display

- "YOUR COSMIC MEMORY SCORE!"
    - Clearly indicates that this screen presents the user's performance outcome.
    - Maintains thematic consistency with the space-based game design.
- Score Message: "YOU GOT 7/9 CORRECT!"
    - Displays the number of correctly placed emojis out of the total.
    - Large, bold yellow text ensures high visibility and instant comprehension.
- Emoji Grid Preview
    - Shows the correctly placed emojis in their grid positions.
    - Helps users visually review their recall performance.

🚀 Feedback Message

- "AMAZING!!!"
    - Provides positive reinforcement and motivation.
    - Encourages user confidence and continued engagement.

🔁 Action Buttons

- "RESTART" Button
    - Allows users to replay the task to improve their score.

    - Supports learning through repetition.
- “EXIT” Button
    - Enables users to end the game session.
    - Provides a clear and simple closure option.

 Thematic Visual Elements

- Space-themed illustrations such as planets, UFOs, and astronauts:
    - Enhance visual engagement without distracting from the score.
    - Maintain a consistent cosmic aesthetic across all frames.

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3. How This Phase Works

1. The system evaluates the user’s emoji placements from the Recall Phase.
2. It calculates the accuracy score based on correct placements.
3. The final score (e.g., 7/9 correct) is displayed prominently.
4. Users interpret their performance using the score and feedback text.
5. Users select:
    - Restart to attempt the task again, or
    - Exit to end the experience.

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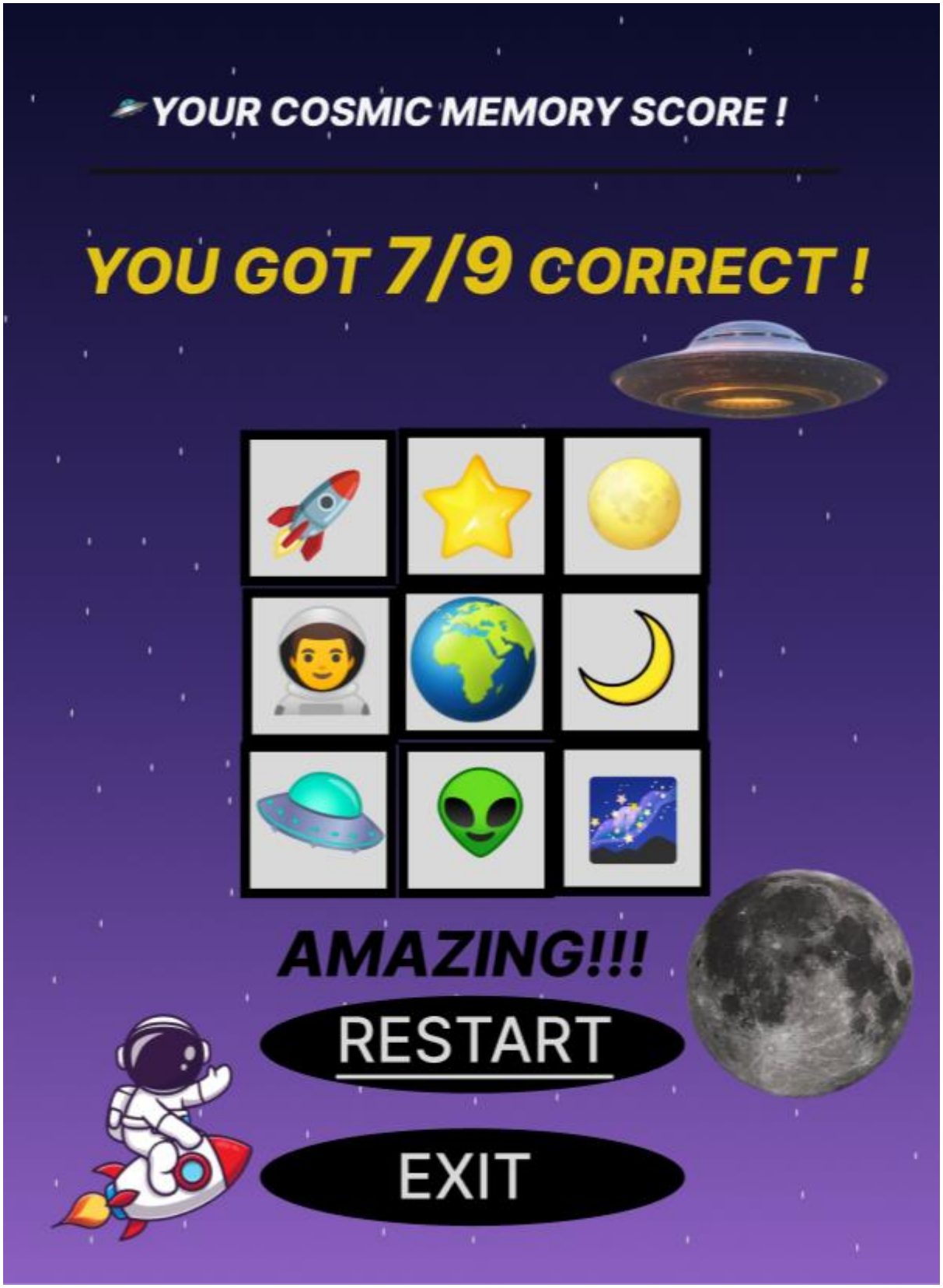
4. Cognitive & UX Benefits

- Immediate feedback strengthens learning and memory awareness.
- Clear score representation helps users self-evaluate performance.
- Positive reinforcement reduces stress and promotes motivation.
- Simple navigation options give users control over their interaction.
- Visual consistency improves usability and overall user experience.



Prototype link:

https://www.figma.com/design/o59YLXtNAitlnQVIlphNlM/Untitled?node-id=0-1&t=IZNlLjXqHq9fUcsR-1